

Correct Statements on Complex Differentiability and Analyticity

Cauchy–Riemann equations, continuity, and harmonic functions

Notation and terminology

Let $\Omega \subseteq \mathbb{C}$ be open, let $z = x + iy$, and write

$$f(z) = u(x, y) + i v(x, y),$$

where $u, v : \Omega \rightarrow \mathbb{R}$. In this note, *analytic* and *holomorphic* are used interchangeably. Also,

$$\Delta u = u_{xx} + u_{yy}$$

denotes the Laplacian of u .

Statement 1: A necessary and sufficient condition for analyticity

The function f is analytic on Ω if and only if

$$u, v \in C^1(\Omega) \quad \text{and} \quad u_x = v_y, \quad u_y = -v_x$$

at every point of Ω . Thus,

$$f \text{ is analytic on } \Omega \iff u, v \in C^1(\Omega) \text{ and satisfy the Cauchy–Riemann equations on } \Omega.$$

Example 1: $f(z) = z^2$

Since

$$z^2 = (x + iy)^2 = (x^2 - y^2) + i(2xy),$$

we have $u = x^2 - y^2$ and $v = 2xy$. Their first partial derivatives are

$$u_x = 2x, \quad u_y = -2y, \quad v_x = 2y, \quad v_y = 2x.$$

Hence $u_x = v_y$ and $u_y = -v_x$ everywhere. Since these partial derivatives are continuous, f is analytic on all of $\mathbb{C}$, and

$$f'(z) = u_x + i v_x = 2x + i(2y) = 2z.$$

Statement 2: Complex differentiability at one point

Let $z_0 = x_0 + iy_0$. The exact pointwise criterion is

$$f \text{ is complex differentiable at } z_0 \iff u \text{ and } v \text{ are differentiable as real functions at } (x_0, y_0), \\ \text{and the Cauchy–Riemann equations hold at } (x_0, y_0).$$

When these conditions hold,

$$f'(z_0) = u_x(x_0, y_0) + i v_x(x_0, y_0) = v_y(x_0, y_0) - i u_y(x_0, y_0).$$

A commonly used sufficient hypothesis for the real differentiability is that the first partial derivatives of u and v exist in a neighborhood of (x_0, y_0) and are continuous at (x_0, y_0) .

Important: the Cauchy–Riemann equations holding only at z_0 are not, by themselves, sufficient.

Counterexample 2: The Cauchy–Riemann equations at one point are not enough

Define $f(0) = 0$, and for $z = x + iy \neq 0$, define

$$f(z) = \frac{x^3}{x^2 + y^2} + i \frac{y^3}{x^2 + y^2}.$$

At $(0, 0)$,

$$u_x = 1, \quad u_y = 0, \quad v_x = 0, \quad v_y = 1,$$

so the Cauchy–Riemann equations hold there. Nevertheless,

$$\frac{f(t) - f(0)}{t} = 1 \quad (t \in \mathbb{R}, t \neq 0),$$

whereas, along $z = t(1 + i)$,

$$f(t(1 + i)) = \frac{t}{2}(1 + i) \implies \frac{f(t(1 + i)) - f(0)}{t(1 + i)} = \frac{1}{2}.$$

The two limits disagree, so $f'(0)$ does not exist.

Statement 3: What it means to be analytic at a point

The function f is analytic at z_0 if and only if there is some $r > 0$ such that $D(z_0, r) \subseteq \Omega$ and f is complex differentiable at every point of $D(z_0, r)$. Equivalently,

$$\begin{aligned} & f \text{ is analytic at } z_0 \\ \iff & f \text{ is complex differentiable throughout} \\ & \text{some open neighborhood of } z_0. \end{aligned}$$

Therefore, complex differentiability only at z_0 does not imply analyticity at z_0 .

Example 3A: Analyticity really concerns a neighborhood

Let

$$f(z) = \frac{1}{z - 1}.$$

At $z_0 = 0$, choose any $r < 1$. The disk $D(0, r)$ does not contain the singular point 1, and f is complex differentiable at every point of that disk. Hence f is analytic at 0, with

$$f'(z) = -\frac{1}{(z - 1)^2}.$$

Counterexample 3B: Differentiable at one point but not analytic there

Let

$$f(z) = |z|^2 = z\bar{z}.$$

At $z_0 = 0$,

$$f'(0) = \lim_{h \rightarrow 0} \frac{|h|^2}{h} = \lim_{h \rightarrow 0} \bar{h} = 0.$$

Thus f is complex differentiable at 0. However, if $f = u + iv$, then $u = x^2 + y^2$ and $v = 0$. The Cauchy–Riemann equations can hold only when $x = y = 0$. Therefore f is not complex differentiable throughout any neighborhood of 0, so it is not analytic at 0.

Statement 4: Complex differentiability implies continuity

If f is complex differentiable at z_0 , then f is continuous at z_0 :

$$f \text{ complex differentiable at } z_0 \implies f \text{ continuous at } z_0.$$

The converse implication is false.

Why Statement 4 holds

For $z \neq z_0$,

$$f(z) - f(z_0) = \frac{f(z) - f(z_0)}{z - z_0} (z - z_0).$$

If $f'(z_0)$ exists, then the first factor tends to $f'(z_0)$, while the second tends to 0. Hence $f(z) - f(z_0) \rightarrow 0$, which is precisely continuity at z_0 .

Counterexample 4: Continuity does not imply complex differentiability

The function

$$f(z) = \bar{z}$$

is continuous everywhere. At any z_0 , however,

$$\frac{f(z_0 + h) - f(z_0)}{h} = \frac{\bar{h}}{h}.$$

Along real h , this quotient equals 1; along purely imaginary h , it equals -1 . Thus the derivative does not exist at any point.

Common mistakes to avoid

- The Cauchy–Riemann equations at one isolated point do not guarantee complex differentiability there.
- Continuity at z_0 does not guarantee complex differentiability at z_0 .
- Complex differentiability at one point does not mean analyticity at that point; analyticity requires a whole neighborhood.
- Harmonicity of u and v does not guarantee that $u + iv$ is analytic; the two functions must be harmonic conjugates.

Statement 5: Analyticity and harmonicity

If $f = u + iv$ is analytic on Ω , then both its real and imaginary parts are harmonic:

$$f \text{ analytic on } \Omega \implies \Delta u = 0 \quad \text{and} \quad \Delta v = 0.$$

The converse is false if we assume only that u and v are harmonic. The correct equivalence is

$$f = u + iv \text{ is analytic} \iff u \text{ and } v \text{ are harmonic conjugates,}$$

where “harmonic conjugates” means that u and v occur together in the Cauchy–Riemann equations

$$u_x = v_y, \quad u_y = -v_x.$$

Example 5A: Harmonic conjugates

For

$$f(z) = e^z = e^x \cos y + i e^x \sin y,$$

we have $u = e^x \cos y$ and $v = e^x \sin y$. A direct computation gives

$$\Delta u = e^x \cos y - e^x \cos y = 0, \quad \Delta v = e^x \sin y - e^x \sin y = 0.$$

Moreover,

$$u_x = e^x \cos y = v_y, \quad u_y = -e^x \sin y = -v_x.$$

Thus u and v are harmonic conjugates, and e^z is analytic.

Counterexample 5B: Two harmonic parts need not form an analytic function

Again consider

$$f(z) = \bar{z} = x - i y.$$

Here $u = x$ and $v = -y$, and both are harmonic:

$$\Delta u = 0, \quad \Delta v = 0.$$

But

$$u_x = 1 \neq -1 = v_y,$$

so the Cauchy–Riemann equations fail. Consequently, f is not analytic.

One-line summary

$$u, v \in C^1(\Omega) \text{ and satisfy CR on } \Omega \iff f \text{ is analytic on } \Omega,$$

$$f \text{ differentiable at } z_0 \implies f \text{ continuous at } z_0,$$

$$f \text{ analytic} \implies u, v \text{ harmonic,}$$

but neither continuity nor harmonicity alone implies complex differentiability or analyticity.